Verifiable Intent-Driven Link Adaptation: A Multi-Timescale Agentic Framework for 6G Physical Layers

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Abstract—Multimodal foundation models can now generate a complete physical-layer (PHY) link configuration (modulation, code rate, power, channel code, estimator, equaliser and precoder) from channel state information and a natural-language intent in a single forward pass. Deployed on a live link, however, such a generator is open-loop: it has no memory of outcomes, no criterion for when to re-plan, and no guarantee that its output can be signalled by a 3GPP stack. We present VILA, a multi-timescale agentic framework that wraps an intent-conditioned generator with three components. A token-budgeted semantic memory records outcomes and is invalidated by cause. A reflection module produces a semantic deviation score, which is fused with CUSUM and Bayesian change-point evidence under a calibrated threshold and a cooldown. A table-driven safety envelope constrains every executed configuration. For the envelope we prove that prefix-completable constrained decoding is sound and complete, and with nearest-feasible projection of held configurations it guarantees admissibility under every trigger history; all 7560 configurations and 14 envelope states are verified exhaustively. We evaluate VILA on an open, time-varying closed-loop simulator built on the AGENS action space, with pre-registered hypotheses, 20 seeds per test set and corrected non-parametric tests. VILA removes all protocol violations at no measurable task cost (pooled equivalence within 2%). Relative to every-cycle generation, it cuts generator invocations by $40\times$ and configuration switching by $15\times$, and relative to fixed-period re-planning it reduces regret after intent changes by 67%. The evaluation also reports what did not work. With a compact backbone, the memory brought no gain, fusing statistical detectors did not beat the semantic score, and every-cycle generation keeps the lowest regret when inference cost and latency are ignored. These results delimit what closed-loop agentic control buys at the PHY and where language-model capacity is still needed.

Index Terms—6G, link adaptation, large language models, agentic AI, intent-based networking, change-point detection, safe learning, constrained decoding.

1 Introduction

SIXTH-generation (6G) air interfaces are expected to be AI-native: instead of fixing the physical-layer (PHY) chain at design time and tuning a handful of parameters with hand-crafted loops, the radio should understand what a service needs and configure itself accordingly [1], [2]. Two lines of work make this plausible. First, foundation models for wireless signals have shown that a pretrained backbone can be adapted to channel prediction, beam prediction and resource allocation with little task-specific data [3], [4], [5], [6], [7], [8], [9], [10]. Second, large language models (LLMs) adapted to networking tasks with modality encoders, task heads and low-rank adaptation (LoRA) now match or exceed specialised learners [11], [12], [13], [14]. AGENS [15] brings these threads to the PHY. It aligns channel state information (CSI) with a natural-language user intent inside a GPT-2 Medium backbone and generates, in a single forward pass, a complete link configuration covering modulation, code rate, transmit-power offset, channel code, channel estimator, equaliser and precoder. Compared with greedy and beam search over the same configuration space, AGENS finds better intent-specific trade-offs and needs orders of magnitude less inference time [15].

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AGENS, like most LLM-for-PHY work, is an *open-loop, single-step* decision maker. Each decision is generated from the current CSI and intent text, independently of what happened before, and nothing checks afterwards whether the decision achieved what the user asked for. Deployed on a real link, such a generator has to keep working while the channel moves between propagation regimes, interference that the CSI report does not show comes and goes, users rephrase or reverse their intents, and network-level constraints such as transmit-power caps change. A single-step generator has three options in that setting, and each is unsatisfactory. It can regenerate every cycle, which pays the full model-inference cost continuously, makes the configuration switch frequently, and still cannot learn from outcomes. It can regenerate on a fixed period, which reacts late to real changes and wastes inference when nothing has changed. Or it can run a classical outer-loop correction [16], [17], which helps with mis-calibrated signal-to-interference-plus-noise ratio (SINR) estimates but does not see intent changes at all. In every case, nothing guarantees that the generated configuration can actually be signalled and executed by a 3GPP stack. The reference implementation, for example, silently aliases unsupported convolutional-code rates, and nothing prevents a power offset above a cap imposed by the network.

This paper asks what it takes to turn an intent-aware single-step generator into a *short-horizon agentic* PHY controller. Such a controller tracks the intent over time and

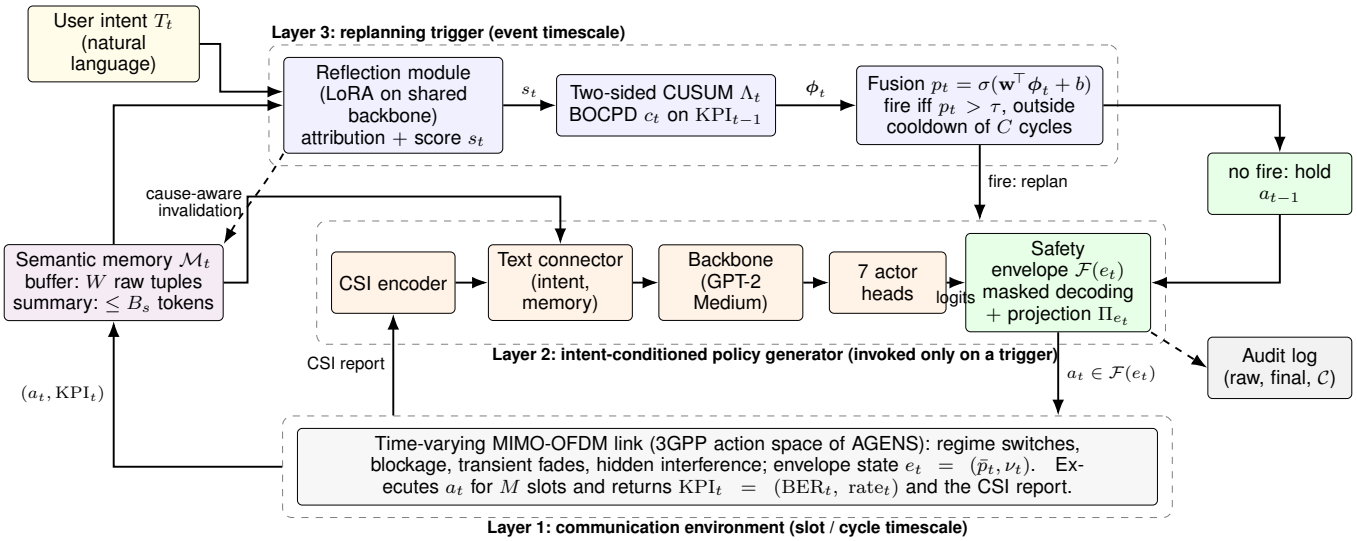


Fig. 1. Architecture of VILA, a three-layer multi-timescale closed loop around an AGENS-style intent-conditioned policy generator. The trigger layer fuses the semantic deviation score of the reflection module with statistical change evidence. The generator is invoked only when the trigger fires, and every executed configuration, newly generated or held, passes through the table-driven safety envelope.

notices when its current configuration has stopped serving that intent. It re-plans when it should, and only then, and it never emits an inadmissible configuration. We organise the problem around three questions:

- **Q1 (memory and consistency).** How can the history of decision–feedback pairs be compressed into a memory that fits the context budget of the generator, so that closed-loop decisions remain consistent while the intent evolves slowly, rather than degenerating into memoryless re-planning at every cycle?
- **Q2 (calibrated failure detection).** Without dense failure labels, how should semantic attribution signals be fused with statistical change-point evidence so that the controller does not over-react to transient fading but does not lag behind genuine distribution shifts or intent changes?
- **Q3 (verifiable safety envelope).** How can PHY realisability constraints be embedded in generation and re-planning so that the output provably stays inside the admissible set under every trigger condition, without relying on the internal representations of the learned model?

We answer them with *VILA* (Verifiable Intent-driven Link Adaptation), a three-layer, multi-timescale closed loop around an AGENS-style generator (Fig. 1). A hierarchical semantic memory keeps the most recent decision–feedback tuples verbatim and folds older history into a token-bounded recursive summary of intent trajectory, regime switch points and recurring failure modes; entries are invalidated when the reflection module attributes a deviation to a regime change. A reflection module produces a cause attribution and a semantic deviation score, which a calibrated logistic fusion rule combines with two-sided cumulative-sum (CUSUM) [18] and Bayesian online change-point detection (BOCPD) [19] statistics; a cooldown bounds the re-planning rate. A safety envelope encodes 3GPP modulation-and-coding consistency [20], codec admissibility, the dynamic power cap and receiver realisability as explicit tables. It enforces them through prefix-completable constrained decoding with a nearest-feasible projection as fallback, and it applies the same projection to *held* configurations when the envelope changes.

The contributions of this paper are as follows.

1) **Framework.** We formulate intent-driven link adaptation over time as a constrained, partially observed control problem with explicit switching and re-planning costs (Section 3). Unlike AGENS and other LLM-for-PHY work, which generate configurations in open loop, VILA embeds an intent-conditioned multimodal PHY policy generator in a short-horizon agentic loop. It has three reusable components: semantic memory compression with cause-aware invalidation, dual-signal failure detection with a cooldown, and a verifiable safety envelope (Section 4).
2) **Guarantees.** We prove that prefix-completable constrained decoding is sound and complete for the admissible set. Together with the projection fallback, this makes every executed configuration admissible for every trigger history. We also bound the switching rate induced by the cooldown and relate the CUSUM threshold to the false-alarm run length. The envelope properties are checked exhaustively over all 7560 raw configurations and 14 envelope states ($231{,}672$ checks, zero failures).
3) **Open closed-loop testbed and pre-registered evaluation.** We build a time-varying closed-loop link simulator that reuses the AGENS action space and reward. It injects regime switches, blockage, transient fades, hidden interference, intent drift, paraphrased restatements and power caps, and provides in-distribution, distribution-shift and adversarial test sets. We test three pre-registered hypotheses (H1–H3) with 20 seeds per set, Wilcoxon signed-rank tests with Holm–Bonferroni correction and

a two-one-sided-test (TOST) equivalence analysis (Sections 5–6).

4) **Quantified extension cost.** We characterise what moving from single-step generation to continuous responsibility costs and buys, in terms of regret, intent satisfaction, switching, re-planning rate, context tokens and backbone compute. We report where the hypotheses hold and where they do not.

The scope of the evidence must be stated up front. The experiments use a link-abstraction surrogate of the AGENS Sionna chain and, because no GPU or model download was available, a compact learned backbone in place of GPT-2 Medium for both the generator and the reflection scorer (Section 5.6). Accordingly, the claims tested here are about the closed-loop *architecture*, namely memory, triggering and the envelope, around a fixed intent-conditioned generator, and not about the language-modelling capacity of a particular LLM. The envelope guarantees do not depend on the backbone at all.

The rest of the paper is organised as follows. Section 2 reviews related work. Section 3 presents the system model and problem formulation. Section 4 details VILA and its analysis. Section 5 describes the experimental methodology, and Section 6 reports the results. Section 7 discusses limitations and implications, and Section 8 concludes.

2 Related Work

2.1 Link Adaptation

Adaptive modulation and coding has a long history, from variable-rate variable-power M-QAM [21] to the channel-quality-indicator-driven modulation and coding scheme (MCS) selection of LTE and NR [20]. Because CSI reports are quantised, delayed and blind to some interference, practical systems add outer-loop link adaptation (OLLA). OLLA adjusts an SINR offset from acknowledgement (ACK)/negative acknowledgement (NACK) feedback so that a block-error-rate target is met [16], and later work self-tunes the OLLA step sizes [17]. Learning-based link adaptation has used contextual bandits [22], reinforcement learning [23], deep-learning predictors for ultra-reliable low-latency communication (URLLC)/enhanced mobile broadband (eMBB) multiplexing [24], offline RL and sequence models [25], and adaptive SINR inference from ACK/NACK [26]. In mobile computing, rate adaptation for 802.11 has been driven by effective-SNR prediction [27] and by optimal rate sampling [28]. All of these methods adapt one or two knobs (usually the MCS) to a fixed objective. None of them conditions jointly on natural-language intent and the whole multi-module PHY chain, and none comes with an admissibility guarantee for the configuration it outputs. VILA keeps OLLA as a baseline and borrows its feedback idea, but generalises the feedback from an SINR offset to semantic memory over complete configurations and intents.

2.2 Foundation Models and LLMs for Wireless and Networking

Pretrained models have been adapted to channel prediction [3], [6], beam prediction [4], resource allocation [5], and multi-task channel representation [7], [8], [9], [10]. NetLLM [11] established a recipe for networking tasks, namely a frozen LLM with multimodal encoders, task-specific heads and LoRA [12], and showed that it generalises better than specialised learners. Surveys and position papers argue for LLM-driven wireless intelligence and agents [13], [14], [29]. AGENS [15] is the closest work and our starting point. It aligns CSI and intent text in GPT-2 Medium [30], uses a Q-Former-style connector to compress the intent tokens, and generates a seven-module link configuration in one step, trained by supervised fine-tuning (SFT) on heuristic-search experience followed by policy-gradient fine-tuning. VILA retains this generator and its interface but changes the problem from a one-shot decision to sustained responsibility over a session. Intent-based networking [31] pursues the same goal at the management plane. VILA can be read as intent-based operation pushed down to the PHY configuration timescale.

2.3 Agentic Memory and Reflection

LLM agents interleave reasoning and acting [32], learn from verbal self-reflection [33], and manage bounded context with hierarchical memory [34] or reflection-based memory streams [35]. These ideas assume tasks measured in seconds to minutes, with feedback in text. The PHY setting adds three constraints that existing agent memories do not address. The context budget is fixed by the connector architecture. Feedback arrives as noisy key performance indicator (KPI) time series rather than text. And part of the remembered history becomes misleading as soon as the propagation regime changes. VILA's memory is designed around these constraints through a strict token budget, structured serialisation and cause-aware invalidation, and its fidelity is monitored with memory probes.

2.4 Change Detection and Concept Drift

CUSUM [18] is optimal in Lorden's minimax sense for detecting a change in distribution [36], [37], and BOCPD [19] maintains a posterior over the run length since the last change. Both are standard tools for abrupt-change detection [38] and for concept-drift adaptation in learning systems [39]. Purely statistical detectors, however, cannot see changes that leave the KPI distribution unchanged at the moment they happen. An intent change is the extreme case: a new intent text alters the objective without altering the channel. Conversely, they cannot tell a transient deep fade from the onset of a blockage without context. VILA fuses statistical evidence with a semantic deviation score and calibrates the fusion on a held-out calibration set.

2.5 Safe Learning-Based Control and Constrained Generation

Shielding [40] and invalid-action masking [41] keep reinforcement-learning agents inside a safe or valid action set, and guided generation constrains LLM decoding to a formal grammar [42]. VILA's envelope applies the same principle to a factorised multi-head PHY action space with cross-module constraints. Its masks are computed from the

TABLE 1
Positioning of VILA against representative prior work.

Work	Intent	Multi-module	Closed loop	Change detection	Verified envelope
OLLA [16], [17]	–	–	✓	–	–
Bandit/RL link adaptation [22], [23]	–	–	✓	–	–
NetLLM [11]	–	–	–	–	–
AGENS [15]	✓	✓	–	–	–
Reflexion, MemGPT [33], [34]	✓	n/a	✓	–	–
VILA (ours)	✓	✓	✓	✓	✓

joint feasibility table so that greedy per-head decoding cannot dead-end, and a projection covers held actions whose admissibility changes over time. Because the admissible set is finite and table-defined, the guarantee can be checked exhaustively instead of argued statistically.

Table 1 summarises how VILA differs from representative prior work.

3 System Model and Problem Formulation

3.1 Link, Configuration Space and Timescales

We consider the downlink of the AGENS reference link [15]: a 4×2 multiple-input multiple-output (MIMO) orthogonal frequency-division multiplexing (OFDM) link with $N_s = 2$ spatial streams, 30 kHz subcarrier spacing, a 128-point FFT with 108 used subcarriers and 14 OFDM symbols per slot, two of which carry Kronecker-pattern pilots [43]. A *configuration* (action) is the seven-tuple used by the AGENS code,

$$a = (a_{\mathrm{mod}}, a_{\mathrm{rate}}, a_{\mathrm{pow}}, a_{\mathrm{code}}, a_{\mathrm{ce}}, a_{\mathrm{eq}}, a_{\mathrm{prec}}) \in \mathcal{A} = \textstyle\prod_{k=1}^{7} \mathcal{A}_k, \tag{1}$$

with modulation $\in$ {QPSK, 16QAM, 64QAM, 256QAM}, code rate $\in \{1/3, 1/2, 2/3, 3/4, 4/5\}$, transmit-power offset $\in \{0, 1, 2, 3, 4, 6, 8\}$ dB, channel code $\in$ {low-density parity-check (LDPC), Polar, convolutional}, channel estimator $\in$ {least squares (LS) with nearest-neighbour interpolation (LS-NN), LS with linear interpolation (LS-Lin), linear minimum mean-square error (LMMSE)}, equaliser $\in$ {LMMSE, matched filter (MF), zero-forcing (ZF)} and precoder $\in$ {ZF, regularised zero-forcing (RZF)}, so that $|\mathcal{A}| = 7560$.

Time is organised on three nested scales. A *slot* (0.5 ms) is the transmission unit. A *cycle* t aggregates $M = 10$ slots (5 ms) and is the decision period: one configuration a_t is applied throughout the cycle, and the cycle yields one KPI observation. Precoders are recomputed every slot from sounding-based CSI with a 2 ms delay, which is the fast PHY loop that VILA does not replace. *Re-planning events*, the invocations of the generator, occur on a third, irregular timescale determined by the trigger. Between events the current configuration is held. The cycle length matches the 10 ms–1 s control range of an O-RAN near-real-time RAN intelligent controller (RIC) [44], which is where we envisage VILA running.

3.2 Observations, Intents and Reward

At the start of cycle t the controller observes

$$o_t = \big(\mathbf{h}^{\mathrm{rep}}_{t-1},\ T_t,\ e_t,\ \mathrm{KPI}_{t-1}\big), \tag{2}$$

where $\mathbf{h}^{\mathrm{rep}}_{t-1}$ is the CSI report of the previous cycle (per-sub-band singular values, a wideband signal-to-noise ratio (SNR) estimate, and delay-spread, Doppler, K-factor and conditioning descriptors), T_t is the current intent text, e_t is the envelope state defined below, and $\mathrm{KPI}_{t-1} = (\mathrm{BER}_{t-1}, \mathrm{rate}_{t-1})$ is the realised outcome of the previous cycle. The CSI report does not show co-channel interference, which therefore acts as a hidden state variable that can only be inferred from outcomes, as in practical OLLA.

Each intent text expresses a latent preference $i_t \in \mathcal{I}$: high throughput, high reliability, energy saving, a balanced fallback for vague requests, and a reliability-plus-low-power combination that never appears in training. Following the AGENS reward, the KPI-mean reward of configuration a in cycle t is

$$\bar{r}_t(a) = \textstyle\sum_{j \in \{\mathrm{ber,rate,time,pow}\}} w_j(i_t)\, R_j\big(\overline{\mathrm{KPI}}_t(a)\big), \tag{3}$$

where R_j are the piecewise normalisations of the AGENS code (`norm_ber`, `norm_rate`, `norm_time`, `norm_power`) and $w_j(i)$ the intent-specific weights listed in Table 3. We make one documented change to the AGENS reward. The bit error rate (BER) reference of `norm_ber` is intent-conditioned: 10^{-5} for reliability, 10^{-4} for reliability-plus-low-power, and 10^{-2} otherwise. Without this change the reward saturates at $\mathrm{BER} \le 10^{-2}$ and cannot express a reliability intent.

3.3 Admissible Configurations and the Envelope State

Not every element of $\mathcal{A}$ can be signalled or executed. The envelope state $e_t = (\bar{p}_t, \nu_t)$ consists of the transmit-power cap index $\bar{p}_t$, which may be lowered by the network for interference coordination or exposure limits, and a flag ν_t indicating whether channel covariance statistics are valid; they are invalid for 5 cycles after a regime change. The admissible set $\mathcal{F}(e_t) \subseteq \mathcal{A}$ consists of the configurations that satisfy four table-defined constraint families:

- **C1** *MCS-table consistency*. The code rate must lie in the rate window that 3GPP TS 38.214 MCS Tables 5.1.3.1-1/2 provide for the selected modulation [20], with tolerance $[-0.01, +0.03]$.
- **C2** *Codec admissibility*. The convolutional code supports only $R \in \{1/3, 1/2\}$, whereas the AGENS code silently aliases other indices.
- **C3** *Power cap*. $a_{\mathrm{pow}} \le \bar{p}_t$.
- **C4** *Receiver realisability*. LMMSE interpolation requires $\nu_t = 1$.

C1 and C2 leave $|\mathcal{F}| = 3906$ of the 7560 raw configurations. With C3 and C4, $|\mathcal{F}(e)|$ ranges from 372 to 3906 over the 14 envelope states.

3.4 Problem Formulation

A closed-loop controller consists of a trigger $g_t \in \{0, 1\}$ and a generator. When $g_t = 1$ the generator produces a new configuration; otherwise the previous one is held. Let $r^\star_t = \max_{a \in \mathcal{F}(e_t)} \bar{r}_t(a)$ be the reward of a clairvoyant oracle

Algorithm 1 VILA closed loop (one cycle t)

Require: $o_t = (\mathbf{h}^{\text{rep}}_{t-1}, T_t, e_t, \text{KPI}_{t-1})$, memory $\mathcal{M}_{t-1}$, previous configuration a_{t-1}
$(\Lambda_t, c_t) \leftarrow$ CUSUM/BOCPD update with $\mathbf{x}_{t-1}$ ▷ statistical evidence
if pre-screen passes (or gating disabled) **then**
  $(s_t, \hat{\kappa}_t) \leftarrow \textsc{Reflect}(a_{t-1}, \text{KPI}_{t-1}, T_t, \mathcal{M}_{t-1})$ ▷ score, cause
else
  $s_t \leftarrow 0$, $\hat{\kappa}_t \leftarrow$ none
end if
$p_t \leftarrow \sigma(\mathbf{w}^\top \boldsymbol{\phi}(s_t, \Lambda_t, c_t) + b)$
if $p_t > \tau$ **and** $t - t_{\text{last}} > C$ **then** ▷ re-plan
  if $\hat{\kappa}_t \in \{\text{regime}, \text{blockage}\}$ **then**
    invalidate buffer of $\mathcal{M}_{t-1}$
  end if
  $\boldsymbol{\ell}_{1:7} \leftarrow \textsc{Generator}(\mathbf{h}^{\text{rep}}_{t-1}, [T_t \| e_t \| \mathcal{S}_{t-1} \| \mathcal{B}_{t-1}])$
  $a_t \leftarrow$ constrained decoding (8) in $\mathcal{F}(e_t)$; reset CUSUM/BOCPD; $t_{\text{last}} \leftarrow t$
else
  $a_t \leftarrow \Pi_{e_t}(a_{t-1})$ ▷ hold; re-projected if e_t changed
end if
log $(a^{\text{raw}}, a_t, \mathcal{C})$ to the audit log if corrected
execute a_t for M slots; observe KPI_t, $\mathbf{h}^{\text{rep}}_t$
$\mathcal{M}_t \leftarrow$ push (t, a_t, KPI_t); absorb evicted tuples into $\mathcal{S}$; enforce budget B_s

that knows the full channel, the hidden interference and the true intent. We seek a controller that solves

$$\begin{aligned} \min\ & \mathbb{E}\Big[\textstyle\sum_{t=1}^{T} \big(r^\star_t - \bar{r}_t(a_t)\big)\Big] + \lambda_{\text{sw}}\, \mathbb{E}\Big[\textstyle\sum_t \mathbb{1}\{a_t \neq a_{t-1}\}\Big] \\ & + \lambda_{\text{c}}\, \mathbb{E}\Big[\textstyle\sum_t g_t\Big] \\ \text{s.t.}\ & a_t \in \mathcal{F}(e_t) \quad \forall t, \text{ surely}, \end{aligned} \tag{4}$$

where the first term is the cumulative regret, the second penalises configuration switching (each switch costs signalling and may disturb hybrid automatic repeat request (HARQ) processes and higher-layer rate control), and the third penalises generator invocations, which dominate the compute cost. The constraint is *sure* rather than in expectation: a single inadmissible configuration is a protocol violation. Problem (4) is a constrained partially observed control problem whose hidden state includes the propagation regime, the interference level and the latent intent. We do not solve it optimally. Instead, VILA decomposes it into a memory that summarises the history relevant to the hidden state (Q1), a trigger that chooses g_t (Q2), and an envelope that enforces the constraint by construction (Q3). The weights λ_{sw} and λ_{c} are not tuned explicitly; we report the three terms separately so that any weighting can be applied afterwards.

4 The VILA Framework

Fig. 1 shows the architecture and Algorithm 1 the per-cycle procedure. The generator (Layer 2) is the AGENS model: a CSI encoder, a text connector that compresses the token embeddings of its textual input by linear projection, self-attention and CSI-queried cross-attention, a GPT-2 Medium backbone, and seven actor heads. VILA changes only what is fed to the connector, which now receives the intent *plus memory*, and what happens to the head logits, which are now decoded inside the envelope. It adds the memory, the reflection module and the trigger around the generator.

4.1 Semantic Memory Compressor (Q1)

Design principle: recent history verbatim, distant history semantic. The memory $\mathcal{M}_t = (\mathcal{B}_t, \mathcal{S}_t)$ has two parts.

Short-term buffer. $\mathcal{B}_t$ holds the last W decision–feedback tuples $(t', a_{t'}, \text{BER}_{t'}, \text{failed slots}, \text{rate}_{t'}, \text{SNR}^{\text{rep}}_{t'})$, each serialised with a fixed template, e.g.,

```
[t=123] act: mod=64QAM, rate=2/3, power=+4dB,
code=LDPC, CE=LMMSE, EQ=LMMSE, prec=RZF | obs:
  BER=3.2e-03 (fail 3/10), rate=2.41 b/RE |
                SNR=18.3 dB
```

Under the GPT-2 byte-level byte-pair encoding (BPE) vocabulary, an entry never exceeds $b_e = 81$ tokens over the whole configuration space.

Long-term summary. When a tuple leaves the buffer, it is absorbed into $\mathcal{S}_t$. This summary records exactly three kinds of content: the intent trajectory (run-length encoded intent labels with start cycles), attributed switch points (cycle and cause), and recurring failure modes (counts of failures under high-order MCS, robust MCS and low power). The summary is regenerated recursively. Whenever its serialisation exceeds the budget B_s, the oldest intent runs are dropped first (keeping the last two), then the oldest switch points (keeping the last), and finally all but the dominant failure mode.

Cause-aware invalidation. When the trigger fires and the reflection module attributes the deviation to a regime switch or a blockage, buffered tuples describe a channel that no longer exists. They are therefore moved into the long-term summary, and the buffer and channel statistics are reset. An intent change, by contrast, keeps the buffer, because the channel evidence it contains is still valid.

Context assembly and budget. The connector input is [intent T_t] $\|$ [constraints e_t] $\|$ $\mathcal{S}_t$ $\|$ $\mathcal{B}_t$, followed by the CSI embedding. Its length obeys

$$L_{\text{ctx}} \leq L_T + L_e + B_s + W b_e + L_0, \tag{5}$$

with $L_T \leq 23$ tokens for every intent text used here, $L_e = 18$ and $L_0 = 5$ tokens of delimiters. For the default $B_s = 150$ and $W = 4$, $L_{\text{ctx}} \leq 520$ tokens, about half of the 1024-token context of GPT-2 Medium, which leaves room for generation. Because the AGENS connector maps a variable number of text tokens onto a fixed number of query tokens, L_{ctx} affects only the embedding and connector cost. The backbone always processes the same number of positions (Section 4.6).

Memory probes. To detect when compression loses information, we query the summary with four probes: the number of intent changes, the previous intent, the cycle of the last regime switch (within ± 2), and the dominant failure mode. If probe accuracy falls below a threshold, the summary is regenerated with a shorter look-back.

4.2 Reflection Module (Q1, Q2)

The reflection module reads $(a_{t-1}, \text{KPI}_{t-1}, T_t, \mathcal{M}_{t-1})$ together with the drift of the CSI descriptors relative to the running statistics of the current memory epoch. It outputs an attribution over five causes (none, regime switch, blockage/SNR drop, intent update, transient fade) and a scalar deviation score $s_t \in (0, 1)$ that measures whether replanning is needed. In the LLM instantiation, it shares the

TABLE 2
Inputs of the reflection module (all computable from o_t and $\mathcal{M}_{t-1}$).

Group	Inputs
Intent	text changed; Jensen–Shannon divergence between the current and remembered intent posteriors
Channel drift	SNR deviation from the epoch mean; absolute drift of delay-spread, Doppler, K-factor and conditioning descriptors; mean drift of the singular-value profile
Outcome	failed-slot fraction and $\log_{10}$BER of the last cycle; failure fraction relative to its running mean; mean failure fraction of the last three tuples
Temporal context	persistence of an SNR drop (> 4 dB); SNR deviation in the previous cycle

frozen GPT-2 Medium backbone with the generator through a separate rank-8 LoRA adapter, so that the two tasks do not interfere. The adapter has a sigmoid head for s_t and decodes a short attribution text. The serialised input follows the template of the buffer entries plus a drift line, e.g., `[Channel drift] delay spread +40%, K-factor −6 dB`. The input features that the module actually consumes are listed in Table 2.

Supervision without dense failure labels. Failures are never labelled by hand. Instead, three kinds of event are injected in simulation: (i) channel-regime switches and sustained blockage, (ii) intent drift, and (iii) transient deep fades of 10–14 dB lasting one or two cycles. A fourth kind, paraphrased restatement of the current intent, is added to the calibration and test sets. Cycles in a three-cycle window after an event of kind (i) or (ii) are positives for s_t ($y_t = 1$). Kinds (iii) and (iv) are negatives, which teaches the module to separate "re-plan" from "keep executing". The cause label of the most recent event provides the attribution target. Because events of kind (i) are only observable from the KPI of the next cycle, their windows start at $e+1$, whereas intent windows start at e. Training rollouts are generated with a behaviour policy that re-plans after a random delay of 1–5 cycles following each event, plus spontaneous re-plans with probability 0.05 per cycle. During development we found that two otherwise natural inputs, the number of cycles since the last re-plan and the age of the memory epoch, leak the behaviour policy's re-planning times into the labels when the re-plans are aligned with the events. With these inputs, the validation area under the ROC curve (AUC) was 0.998, yet regime switches went undetected on the calibration set. We therefore excluded both inputs (Table 2); the validation AUC of the retained model is 0.888, with a cause accuracy of 94%.

4.3 Replanning Trigger (Q2)

Statistical signal. Let $\mathbf{x}_t = (\log_{10}(\mathrm{BER}_t + 10^{-6}), \mathrm{rate}_t, \mathrm{SNR}_t^{\mathrm{rep}})$. For each component j we run a two-sided CUSUM on the standardised innovations $z_{t,j} = (x_{t,j} - \hat{\mu}_{t,j})/\sigma_j$:

$$S_{t,j}^{+} = \max\big(0, S_{t-1,j}^{+} + z_{t,j} - k\big), \\ S_{t,j}^{-} = \max\big(0, S_{t-1,j}^{-} - z_{t,j} - k\big), \tag{6}$$

and set $\Lambda_t = \max_j \max(S_{t,j}^{+}, S_{t,j}^{-})/h$. Innovations are winsorised at ± 4 to limit the influence of the heavy-tailed BER stream. The reference mean $\hat{\mu}$ is re-estimated over a two-cycle warm-up after every re-plan, because a new configuration legitimately shifts the KPI distribution, and it then tracks the stream with forgetting factor λ_μ. A positive λ_μ makes the drift tolerance adaptive, so that slow shadowing is absorbed rather than accumulated. The scales σ_j are the standard deviations of event-free first differences in the calibration set divided by $\sqrt{2}$. The triple (λ_μ, k, h) is chosen by a grid scan, $\lambda_\mu \in \{0.02, 0.1, 0.3\}$, $k \in \{0.25, \ldots, 3\}$ and $h \in \{2, \ldots, 20\}$, that maximises event-level F_1 subject to a false-trigger rate of at most 5%. The scan is run on calibration rollouts in which the generator re-plans on a fixed 20-cycle schedule, so that the statistics evolve through the events as they would before a trigger fires. As a second statistical signal, BOCPD [19] with a Gaussian likelihood, a conjugate prior on the mean and hazard $1/100$ is run on $\mathrm{SNR}^{\mathrm{rep}}$. Its output is $c_t = P(r_t \le 2 \mid x_{1:t})$, the posterior probability of a change within the last two cycles; it is more sensitive than CUSUM to gradual drifts.

Fusion and calibration. The trigger probability is

$$p_t = \sigma\big(\mathbf{w}^{\top}\boldsymbol{\phi}_t + b\big), \qquad \boldsymbol{\phi}_t = \big(\operatorname{logit} s_t,\ \Lambda_t,\ c_t,\ s_t\Lambda_t\big), \tag{7}$$

and $g_t = \mathbb{1}\{p_t > \tau\} \cdot \mathbb{1}\{t - t_{\mathrm{last}} > C\}$. The weights $(\mathbf{w}, b)$ are fitted by class-balanced logistic regression on the calibration logs. These logs pool the fixed-schedule rollouts with one round of on-policy rollouts of the trigger itself, because the detector statistics depend on when the trigger has reset them. The threshold τ is then chosen by a closed-loop sweep over $\tau \in [0.30, 0.98]$ on the calibration set, maximising event-level F_1 subject to a false-trigger rate of at most 5%. Sweeping in closed loop matters because each trigger resets the detectors and the cooldown, so offline threshold selection on logged scores is biased. The single-signal baselines use the same procedure restricted to the corresponding coordinates of $\boldsymbol{\phi}_t$. The cooldown forces at least $C = 3$ cycles of execution after each trigger.

Proposition 1 (Switching and re-planning rate). With cooldown C, any window of L consecutive cycles contains at most $\lceil L/(C+1) \rceil$ generator invocations and at most $\lceil L/(C+1) \rceil + n_e(L)$ configuration switches, where $n_e(L)$ is the number of envelope-state changes in the window.

Proof: Two invocations are separated by more than C cycles, so at most one occurs in each block of $C+1$ consecutive cycles. A held configuration changes only when the projection of Section 4.4 modifies it. Since Π_e is idempotent on $\mathcal{F}(e)$ (Proposition 3), this can happen only in a cycle in which e_t changes. □

Proposition 2 (False-alarm run length). Assume that, between changes, each standardised stream $z_{t,j}$ is i.i.d. $\mathcal{N}(0,1)$. Then each one-sided statistic of the CUSUM above coincides with Page's log-likelihood-ratio CUSUM for a mean shift of $\pm 2k$, scaled by $1/(2k)$. Its average run length to a false alarm is therefore at least e^{2kh}, and that of the trigger $\Lambda_t \ge 1$ over the six one-sided statistics is at least $e^{2kh}/6$.

Proof: For a shift $\delta = 2k$, the log-likelihood-ratio increment is $\delta z - \delta^2/2 = 2k(z-k)$, so the LLR-CUSUM equals

$2k$ S^+ and crossing h by S^+ is equivalent to crossing $2kh$ by the LLR-CUSUM. Lorden's bound [36] gives an average run length of at least e^{2kh} for a single one-sided statistic. The union bound over six statistics yields the second claim. □
Proposition 2 assumes independence and known scales, which hold only approximately for KPI series under a fixed configuration. It is used to sanity-check the calibrated (k, h), not as an operating guarantee. The calibrated false-trigger rate is the operative quantity.

4.4 Verifiable Safety Envelope (Q3)

Design principle: verifiable, auditable, and independent of the learned model's internal representation. The admissible set is defined by the explicit tables C1–C4 and is materialised as a Boolean tensor $\mathbf{F}_e \in \{0,1\}^{4\times5\times7\times3\times3\times3\times2}$ for each envelope state e.

Prefix-completable constrained decoding. The actor heads emit logits $\boldsymbol{\ell}_1, \ldots, \boldsymbol{\ell}_7$. They are decoded sequentially in the order code, modulation, rate, power, estimator, equaliser, precoder. When head k is decoded after the prefix $a_{<k}$ has been fixed, value v is admitted only if *some* completion of $(a_{<k}, v)$ is admissible:

$$m_k(v \mid a_{<k}) = \bigvee_{a_{>k}} \mathbf{F}_e\big[a_{<k}, v, a_{>k}\big], \qquad a_k = \arg\max_{v:\, m_k(v)=1} \ \ell_{k,v}. \tag{8}$$

Inadmissible values receive $-\infty$ logits, so sampling is also possible.

Theorem 1 (Soundness and completeness of constrained decoding). If $\mathcal{F}(e) \neq \emptyset$, then for any logits the decoder (8) returns $a \in \mathcal{F}(e)$, by greedy selection or by sampling. Conversely, every $a \in \mathcal{F}(e)$ is returned for some logits.

Proof: Soundness. We show by induction that the invariant "the prefix $a_{<k}$ has at least one admissible completion" holds for every k. It holds for $k = 1$ because $\mathcal{F}(e) \neq \emptyset$. If it holds for $a_{<k}$, then $m_k(\cdot \mid a_{<k})$ has at least one nonzero entry, and any value with $m_k(v) = 1$ preserves the invariant by the definition (8). After the seventh head the completion is the action itself, which is therefore admissible. *Completeness.* For $a \in \mathcal{F}(e)$, choose $\boldsymbol{\ell}_k$ with its unique maximum at a_k. Since a itself completes every one of its prefixes, $m_k(a_k \mid a_{<k}) = 1$ for all k, so the decoder returns a. □

Masking each head independently would not have this property: per-head masks cannot express the coupling between code, modulation and rate in C1 and C2, and an independently decoded prefix such as (Conv, 64QAM) has no admissible rate.

Nearest-feasible projection. Some configurations bypass the decoder: held configurations whose admissibility changes when e_t changes, and any output of an external or legacy generator. These are mapped by

$$\Pi_e(a) = \arg\min_{a' \in \mathcal{F}(e)} \ d(a, a'), \qquad d(a, a') = \sum_{k\in\mathcal{O}} \omega_k |a_k - a'_k| + \sum_{k\notin\mathcal{O}} \omega_k \mathbb{1}\{a_k \neq a'_k\}, \tag{9}$$

where the ordinal heads $\mathcal{O}$ are modulation ($\omega = 1$), rate (0.5 per index) and power (0.25 per level), and the categorical heads are code (1), estimator, equaliser and precoder (0.5 each). Ties are broken toward lower spectral efficiency, then lower power, which is the conservative direction.

Proposition 3 (Projection). For every $a \in \mathcal{A}$ and every e: $\Pi_e(a) \in \mathcal{F}(e)$; $\Pi_e(a) = a$ if and only if $a \in \mathcal{F}(e)$; and $d(a, \Pi_e(a)) = \min_{a' \in \mathcal{F}(e)} d(a, a')$.

The proof is immediate from (9) and the tie-breaking rule. In VILA, every executed configuration is either the output of (8) or a projected held configuration. Theorem 1 and Proposition 3 therefore give the following corollary.

Corollary 1. For any trigger sequence $\{g_t\}$, any memory content and any generator parameters, the executed configuration satisfies $a_t \in \mathcal{F}(e_t)$ for all t.

Exhaustive verification. Because $\mathcal{A}$ and the set of envelope states are finite, Theorem 1 and Proposition 3 can also be checked mechanically. For each of the 14 envelope states and each of the 7560 raw configurations we verify admissibility, idempotence and minimality of Π_e against brute-force search (105,840 checks). We also verify admissibility of the decoder under one-hot logits on every raw configuration and under 20,000 random logit draws with greedy or sampled decoding (125,832 checks). All checks pass, and the procedure is a unit test in the released code. The audit log records, for every projected or corrected configuration, the raw action, the executed action and the violated constraint identifiers C1–C4.

4.5 Generator Training

The generator is trained in the two stages of AGENS. Stage 1 is SFT with a per-head cross-entropy loss towards the oracle configuration. Stage 2 is policy-gradient fine-tuning. Because the simulator returns the reward of all 7560 configurations of a training state, stage 2 maximises the *exact* expected reward of the factorised policy, $J(\theta) = \mathbb{E}_s\big[\sum_a \prod_k \pi_{\theta,k}(a_k \mid s)\, \tilde{r}_s(a)\big]$. Here $\tilde{r}_s(a) = \bar{r}_s(a)$ for admissible a and -1 otherwise. A small entropy bonus is added, and no score-function variance is incurred. Training states are short sequences ($W + 1$ cycles) from the five training regimes and the three AGENS intent classes. The behaviour policy perturbs the oracle, holds it with probability 0.5, follows the oracle of a different intent with probability 0.3, and keeps fewer than W valid tuples with probability 0.3 to imitate invalidation. In this way the memory features presented to the generator cover the situations met in closed loop. The memoryless generator used by the AGENS baselines is trained on the same data without the memory input.

4.6 Complexity

Per cycle, VILA always runs the CUSUM and BOCPD updates, which cost $O(1)$ and $O(R_{\max})$ with $R_{\max} = 120$, and the reflection module. The generator and the envelope run only when $g_t = 1$. Constrained decoding costs $O(\sum_k |\mathcal{A}_k| \cdot |\mathcal{A}|)$ Boolean reductions in the generic implementation, which is under one millisecond for $|\mathcal{A}| = 7560$, and the projection costs $O(|\mathcal{A}|)$. For the LLM instantiation, the backbone cost of one forward pass over n positions of GPT-2 Medium ($n_L = 24$ layers, width $d = 1024$, $P \approx 302$M non-embedding parameters) is approximately $2Pn + 4n_L n^2 d$ floating-point operations (FLOPs). The generator processes a fixed number of positions $2L$ (CSI tokens plus connector queries). The reflection module processes its

prompt of L_{ctx} tokens and decodes n_g attribution tokens. The amortised backbone cost per cycle is therefore

$$\bar{\mathcal{C}} = \rho_g \, \mathcal{C}_{\mathrm{fwd}}(2L) + \rho_r \left[\mathcal{C}_{\mathrm{fwd}}(L_{\mathrm{ctx}}) + n_g \, \mathcal{C}_{\mathrm{dec}} \right], \qquad (10)$$

where ρ_g and ρ_r are the per-cycle invocation rates of the generator and the reflection module. AGENS-Open has $\rho_g = 1$ and $\rho_r = 0$. Because the reflection prompt is longer than the generator input, gating the reflection module with a cheap pre-screen is the main lever on $\bar{\mathcal{C}}$, as evaluated in Section 6.5. The pre-screen fires on an intent-text change, an SNR deviation above 3 dB, a failed slot, a descriptor drift above a threshold, or $\Lambda_t > 0.5$.

5 Experimental Methodology

5.1 Closed-Loop Link Simulator

AGENS evaluates each configuration with a full Sionna link run [45], which takes seconds per configuration and makes exhaustive oracles and multi-seed closed-loop studies impractical. We therefore built a vectorised link-abstraction surrogate of the same chain. In each cycle it returns the KPIs of *all* 7560 configurations, which gives exact oracles, exact policy gradients and common random numbers across methods. The surrogate is released with the paper, and its parameters are listed in Table 3.

Channel. A six-tap delay-line MIMO channel has an exponential power-delay profile, Kronecker spatial correlation and an optional Rician LOS component with random angles. It is sampled on 12 sub-bands spanning the 108 used subcarriers. The taps evolve between cycles as a first-order autoregressive (AR(1)) process with coefficient $J_0(2\pi f_D T_c)$, where $T_c = 5$ ms. The precoder uses CSI that is 2 ms old, i.e., correlated with coefficient $J_0(2\pi f_D \cdot 2\,\mathrm{ms})$. Eight propagation regimes are defined, named after the 3GPP scenarios [46] whose delay spread, Doppler and K-factor ranges they loosely follow; AGENS itself draws its CSI from Sionna RT [47]. Five are used for training and test set A: urban-micro (UMi) line-of-sight (LOS) pedestrian, UMi non-LOS (NLOS) pedestrian, urban-macro (UMa) NLOS vehicular, indoor-hotspot LOS and an interference-limited cell edge. Three are reserved for test set B: high-speed LOS at 120 km/h, a dense rich-scattering regime with a 500 ns delay spread and low SNR, and a near-rank-one LOS regime. Large-scale SNR follows an AR(1) shadowing process ($\sigma = 2$ dB, coefficient 0.95 per cycle). The hidden interference-over-noise ratio follows a regime-dependent level (0–6 dB) with AR(1) fluctuations.

PHY abstraction. ZF and RZF ($\alpha = 2$) precoders are computed from the delayed CSI and normalised per stream. Pilot-based estimation error combines a noise term, scaled down by a smoothing gain for LMMSE, with time- and frequency-interpolation terms that grow with Doppler and delay spread and are smaller for linear and LMMSE interpolation. LMMSE with stale covariance incurs a mismatch penalty. The post-equaliser SINR of LMMSE, MF and ZF receivers is computed from the true effective channel, the mismatched receive filter, the noise plus hidden interference, and a -28 dB hardware error vector magnitude (EVM) floor. Per-stream, per-sub-band SINRs are mapped to an effective SINR by exponential effective SINR mapping (EESM) [48] with modulation-specific β. Block-error rates follow code-specific logistic curves around the Shannon threshold of the spectral efficiency. The curves include code-specific gaps (LDPC 1.2 dB, Polar with successive cancellation (SC) decoding 2.2 dB, convolutional 2.8 dB), a 0.15 dB-per-bit bit-interleaved coded modulation (BICM) loss and a 0.4 dB puncturing loss at $R = 4/5$. Rate is reported as successfully delivered information bits per resource element, using the AGENS normalisation. Execution time is a deterministic proxy of decoder and estimator cost, and the power term is $\ln(P_{\mathrm{tx}}/P_{\mathrm{ref}})$, as in AGENS. In every cycle, the slot errors and the measurement noise are drawn once and shared by all methods.

TABLE 3
Simulation and training parameters.

Parameter	Value
Antennas / streams	$N_t = 4$, $N_r = 2$, $N_s = 2$
Numerology	30 kHz SCS, FFT 128, 108 used SC, 14 symbols, pilots on 2, 9
Cycle / slots per cycle	$T_c = 5$ ms, $M = 10$
Carrier, CSI delay	3.5 GHz; precoder CSI delay 2 ms
Configuration space	$\lvert\mathcal{A}\rvert = 7560$; $\lvert\mathcal{F}\rvert = 3906$ under C1–C2
Reward weights $(w_{\mathrm{ber}}, w_{\mathrm{rate}}, w_{\mathrm{time}}, w_{\mathrm{pow}})$	throughput $(.05, .8, .1, .05)$; reliability $(.8, .05, .1, .05)$; energy $(.1, .05, .05, .8)$; balanced $(.4, .3, .15, .15)$; reliability+low power $(.45, .05, .05, .45)$ (unseen)
Memory	$W = 4$, $B_s = 150$ tokens, budget 1024
Trigger	cooldown $C = 3$; BOCPD hazard $1/100$
Generator (experiments)	multilayer perceptron (MLP) 3×256, 7 heads; SFT 30 + RL 15 epochs
Reflection (experiments)	MLP 2×64, score + 5-way cause head
Training data	24,000 sequences (generator); 60 rollouts $\times$ 150 cycles (reflection)
Calibration	60 sequences $\times$ 100 cycles
Test sets	A, B, C: 20 seeds $\times$ 300 cycles each

Events. Episodes last $T = 300$ cycles (1.5 s). Each contains about four regime switches, four intent changes, five transient fades (10–14 dB for one or two cycles), a sustained blockage (-10 dB with loss of the LOS component) with probability 0.6, and one paraphrased restatement of the current intent. Events are separated by at least 10 cycles.

5.2 Datasets and Test Sets

The *training set* uses static regimes and fixed intents drawn from the AGENS intent templates (four texts for each of throughput, reliability and energy saving). It trains the generator and, through 60 event-injected rollouts of 150 cycles, the reflection module. The *calibration set* (60 sequences of 100 cycles with injected events) is used only for the CUSUM scan, the fusion weights and the thresholds. Three test sets with disjoint seeds follow.

- **Test set A** (in-distribution): training regimes and AGENS intent texts.
- **Test set B** (distribution shift): the three unseen regimes plus two training regimes, SNR shifted by -3 dB, held-out intent paraphrases, and the unseen reliability-plus-low-power intent.
- **Test set C** (adversarial): vague intents ("Do whatever is best."), an intent *reversal* six cycles after a drift, a regime switch coinciding with an intent change,

paraphrased restatements, and two periods in which the power cap is lowered to 1–3 dB.

5.3 Compared Methods

All methods share the simulator, the CSI report and the trained generators, and differ only in the components listed below. AGENS-Open, the original method, has no envelope. When it outputs an inadmissible configuration, the reference stack's behaviour is emulated: a fallback MCS for C1, silent rate aliasing for C2, power amplifier (PA) clipping for C3, and stale-covariance execution for C4. Each such event is counted as a violation.

- **AGENS-Open**: memoryless generator invoked every cycle.
- **AGENS-Open+Env**: AGENS-Open with the envelope (added to separate the envelope's effect).
- **AGENS-OLLA**: AGENS-Open+Env whose SNR input is corrected by a classical OLLA offset (block error rate (BLER) target 10%, 0.5 dB up-step) [16].
- **AGENS-Fixed**: memoryless generator re-planned every $N = 10$ cycles, with envelope.
- **AGENS-StatOnly** / **AGENS-SemOnly**: memory generator with a trigger calibrated on statistical (Λ_t, c_t) or semantic (s_t) evidence only.
- **AGENS-Stat+Rule**: StatOnly plus a rule that fires whenever the intent string changes. This baseline was not in the original plan and was added to test whether the semantic signal is more than a text-diff detector.
- **Greedy-Replan** / **Beam-Replan**: the AGENS heuristic baselines (coordinate-wise greedy search and beam search of width 2 over the seven modules), driven by the same fused trigger. They evaluate candidates with a nominal link model built from the last CSI report without knowledge of the hidden interference, and search only admissible configurations.
- **VILA**: full system (memory generator, fused trigger, envelope).
- **Oracle**: per-cycle exhaustive search with full state knowledge, which defines the regret.

The ablations remove the memory input of the generator (*VILA-noMem*), replace the trigger by a fixed period of 10 cycles (*VILA-Fixed*), remove the cooldown (*VILA-noCooldown*) or the cause-aware invalidation (*VILA-noInvalidate*), gate the reflection module with the pre-screen (*VILA-Gated*), and replace constrained decoding by post-hoc filtering, i.e., the most probable admissible joint configuration (*VILA-PostHoc*), by unconstrained decoding plus projection (*VILA-Project*), or by no envelope (*VILA-Unguarded*).

5.4 Metrics

Task level: cumulative regret $\sum_t (r_t^\star - \bar{r}_t(a_t))$, cumulative reward and intent satisfaction rate. A cycle is *satisfied* when its KPI lies in an achievability-aware tolerance band of the current intent. For throughput, $\mathrm{BER} \le 10^{-2}$ and a rate of at least 85% of the best rate achievable at $\mathrm{BER} \le 10^{-2}$. For reliability, $\mathrm{BER} \le \max(10^{-5}, 10\,\mathrm{BER}_{\min})$. For energy saving, $\mathrm{BER} \le 10^{-2}$ and a power within 1 dB of the minimum that achieves it. For the balanced intent, a reward within 0.05 of the oracle. For reliability-plus-low-power, the reliability band at 10^{-4} together with the power condition. **Closed-loop behaviour**: switches and re-plans per 100 cycles; event-level precision, recall and F_1 of the triggers, where a positive event (regime switch, blockage, intent change) counts as detected if a trigger occurs within 5 cycles and a trigger is a true positive if it falls in such a window; the false-trigger rate (FPR), i.e., the fraction of event-free cycles with a trigger; and the recovery delay, i.e., the number of cycles from an intent change until the first satisfied cycle (capped at 30). **Safety**: executed-violation rate, raw-infeasibility rate of the generator, and number of projections. **Cost**: generator and reflection invocations per 100 cycles, context tokens, and backbone FLOPs from (10).

5.5 Statistical Protocol

The hypotheses, margins and tests were fixed before the test sets were run. Every method is evaluated on the same 20 independent episodes per test set, with common random numbers. We report means with 95% Student-t confidence intervals over seeds. Methods are compared with VILA by two-sided Wilcoxon signed-rank tests [49] on per-episode cumulative regret, and Holm–Bonferroni correction [50] is applied over all baselines within a test set. The hypotheses are operationalised as follows. **H1**: on the intent-drift windows (the 20 cycles following each intent change) of test set A, VILA reduces cumulative regret by at least 25% relative to AGENS-Fixed, and its switching frequency is not significantly higher (one-sided Wilcoxon, $\alpha = 0.05$). The whole-episode regret is reported alongside. **H2**: on test sets A and B pooled, the event-level F_1 of the fused trigger exceeds that of the better single-signal trigger by at least 15% (relative), with an FPR below 5%. **H3**: constrained decoding does not degrade the task reward relative to the unconstrained-then-filtered variant (VILA-PostHoc). This is tested with paired TOST [51], [52] on per-episode mean reward, margin $\pm 2\%$ of the comparator mean, $\alpha = 0.05$.

5.6 Implementation and Reproducibility

No GPU and no access to model repositories were available in the environment that produced the results. The generator and the reflection module were therefore instantiated with compact networks that keep the interfaces of the LLM design (Table 3). The generator maps CSI features, a three-class intent posterior, the envelope state and a numeric view of $\mathcal{M}_t$ to seven heads. The intent posterior comes from a hashed bag-of-words encoder trained on the AGENS templates, which reaches 67% accuracy on the held-out paraphrases. The reflection module maps the inputs of Table 2 to s_t and a cause. This instantiation corresponds to the "classification head" fallback anticipated in the project plan. Its implication is that the reported numbers measure the value of the closed-loop architecture around a fixed generator, not the language understanding of GPT-2 Medium. Everything else is implemented as described: memory serialisation and token budgets use the exact GPT-2 BPE vocabulary, and the triggers, envelope, statistics and simulator are complete. The training and calibration logs, the envelope verification report and the scripts that regenerate every table and figure are released together with the simulator.

6 RESULTS

Table 4 reports the main comparison, Table 5 the ablations, and Table 7 the outcome of each pre-registered hypothesis. Unless stated otherwise, intervals are 95% confidence intervals over the 20 episodes of a test set, and p-values are Holm-adjusted two-sided Wilcoxon signed-rank tests against VILA.

6.1 E1: Closed-Loop Stability and Intent Tracking (H1)

Event-triggered re-planning trades a modest regret increase for a large reduction in compute and switching. On test set A, VILA invokes the generator 2.5 times per 100 cycles and switches configuration 2.7 times. AGENS-Open regenerates every cycle and switches 40.5 times. The price is regret: 30.9 ± 7.1 for VILA against 13.1 ± 2.3 for AGENS-Open ($p < 10^{-4}$). AGENS-Open tracks the cycle-to-cycle fluctuations of fading and shadowing, which a held configuration by definition cannot. Fig. 2 places the main methods in this regret–invocation plane. VILA and SemOnly occupy the low-invocation end. AGENS-Fixed spends four times as many invocations (9.7 per 100 cycles) and switches 2.7 times as often (7.4) as VILA, without lower whole-episode regret (31.4 ± 5.1; $p = 1.0$). Greedy- and Beam-Replan, driven by the same trigger, roughly double the regret on A (59.5), because their nominal link model cannot see the hidden interference. StatOnly is worst (84.4): the statistical trigger misses almost all intent changes (Section 6.2).

Intent drift. Fig. 3 aligns the per-cycle regret on the 75 intent changes of test set A that have a complete $[-3, +20]$-cycle window (of 80 in total). AGENS-Fixed incurs a regret spike that decays only when the next scheduled re-plan arrives, about five cycles later on average. VILA re-plans in the cycle of the change, because the intent text is part of the observation, and its regret stays at the level of AGENS-Open. Over the 20 cycles after each intent change, VILA's cumulative regret is 66.8% lower than that of AGENS-Fixed (bootstrap 95% CI 56.8–75.3%; one-sided Wilcoxon $p < 10^{-5}$), and its switching rate is lower, not higher. H1 is therefore supported *as operationalised*. The ablations, however, locate the gain. Replacing the memory generator by the memoryless one (VILA-noMem) changes intent-window regret by only 9% ($p = 0.84$), whereas replacing the trigger by a fixed period (VILA-Fixed) raises it by a factor of three ($p < 10^{-5}$). *The H1 effect comes from event-triggered re-planning, not from the semantic memory.* Over whole episodes, VILA and AGENS-Fixed are statistically indistinguishable on A (1.5% difference).

Memory (Q1) yielded no measurable benefit in this setting. In every test set, VILA-noMem attains slightly lower regret than VILA (28.3 vs. 30.9 on A, 92.4 vs. 102.4 on B, 52.4 vs. 59.0 on C), although no difference survives Holm correction. The validation objective tells the same story: the memory-conditioned generator reaches an expected reward of 0.837, against 0.848 for the memoryless one and 0.878 for the oracle. Its 40 additional inputs did not carry enough information about the hidden interference to compensate for the extra variance they introduce. Cause-aware invalidation did not help either. VILA-noInvalidate is nominally better on B and C (91.9 and 45.5); without invalidation the buffer still holds post-switch evidence when a re-plan follows a misattributed event. We regard these as genuine negative results for the numeric memory view used here, and Section 7 discusses what they do and do not imply for an LLM backbone.

6.2 E2: Failure-Detection Calibration (H2)

Fig. 4 decomposes trigger quality. On test sets A and B pooled, the fused trigger reaches precision 0.89, recall 0.73 and $F_1 = 0.80$ at a false-trigger rate of 0.30%. The semantic-only trigger reaches $F_1 = 0.82$ (precision 0.94, recall 0.72, FPR 0.17%), and the statistical-only trigger $F_1 = 0.40$ (FPR 1.2%). Fusion is therefore 2.2% *worse* than the better single signal, with paired per-episode F_1 $p = 0.03$, and **H2 is not supported**. The per-event breakdown explains why. The statistical detectors catch 100% of blockages and 58% of regime switches, but essentially no intent changes (0.6%), and they fire on 55% of transient fades. The semantic score already detects 81% of regime switches and 77% of blockages while firing on only 2% of fades. The calibrated fusion consequently puts little weight on CUSUM ($w_\Lambda = 0.09$ against $w_s = 0.91$ and $w_c = 1.91$ for BOCPD). The statistical evidence buys a higher blockage recall (86% vs. 77%) at the price of more firing on fades (7% vs. 2%).

Is the semantic signal more than a text-diff detector? AGENS-Stat+Rule fires on every intent-string change. It detects all intent changes but also fires on 100% of paraphrased restatements, so its precision drops to 0.63 and its F_1 to 0.70. The learned semantic score fires on 10–15% of restatements. Its weakness is the converse: under held-out paraphrases (set B) it detects only about a quarter of intent changes (24–26%). The bag-of-words intent encoder does not move its posterior for unseen phrasings, so a genuine change looks like a restatement. This is precisely where a language-model backbone is expected to help, and it is the main reason that recall falls from 0.84 on A to 0.61 on B and 0.47 on C.

Sensitivity. Scaling the CUSUM threshold h and the fusion threshold τ by $\pm 20\%$ (Fig. 5) changes the F_1 on test set A between 0.79 and 0.87. The result is insensitive to h, consistent with the small CUSUM weight, and degrades smoothly as τ approaches 1. The calibrated CUSUM alone violates the assumptions of Proposition 2: with $(k, h) = (2, 2)$ the bound predicts a false-alarm run length of at least 497 cycles, whereas the observed false-trigger rate of 4.5% corresponds to roughly 22 cycles. The heavy-tailed, autocorrelated KPI streams are far from the i.i.d. Gaussian model, which is why the operating point has to be calibrated empirically. Removing the cooldown increases switching (3.0 vs. 2.7 on A) and the false-trigger rate (0.5% vs. 0.3% on A+B), and on test set B it raises regret from 102.4 to 112.2 (unadjusted $p = 0.009$, Holm $p = 0.07$).

6.3 E3: Robustness to Distribution Shift

All learned methods degrade on test set B. VILA's per-cycle regret rises from 0.103 to 0.341 (3.3×), AGENS-Open's from 0.044 to 0.236 (5.4×) and AGENS-Fixed's from 0.105 to 0.272 (2.6×). The model-based search baselines degrade least (1.25×), and on B their regret (75) is below VILA's (102.4, Holm $p = 0.12$–0.18). Their search re-evaluates the link physically instead of extrapolating a learned map, but

TABLE 4
Main results over 20 episodes of 300 cycles per test set: cumulative regret (mean ± 95% confidence interval (CI)), intent-satisfaction rate (%), configuration switches and generator invocations per 100 cycles. *: regret differs from VILA (Wilcoxon signed-rank, Holm-adjusted $p < 0.05$).

	Test set A (in-distribution)				Test set B (distribution shift)				Test set C (adversarial)			
Method	Regret	Sat.	Sw.	Gen.	Regret	Sat.	Sw.	Gen.	Regret	Sat.	Sw.	Gen.
AGENS-Open	13.1 ± 2.3*	78.5	40.5	100.0	70.8 ± 11.8*	48.0	57.9	100.0	33.4 ± 7.7*	62.7	54.5	100.0
AGENS-Open+Env	13.1 ± 2.3*	78.5	40.5	100.0	70.8 ± 11.8*	48.0	57.9	100.0	33.3 ± 7.7*	62.7	54.5	100.0
AGENS-OLLA	21.3 ± 3.7*	71.3	41.6	100.0	92.0 ± 20.9	43.5	60.0	100.0	37.9 ± 9.5	60.1	53.8	100.0
AGENS-Fixed	31.4 ± 5.1	69.7	7.4	9.7	81.7 ± 14.0*	43.4	8.2	9.7	43.0 ± 9.5	58.4	7.8	9.7
AGENS-StatOnly	84.4 ± 21.2*	51.4	2.2	2.0	149.8 ± 22.1*	26.8	2.2	2.0	76.8 ± 26.0	43.7	2.3	1.8
AGENS-Stat+Rule	32.0 ± 7.1	65.6	3.7	3.6	104.9 ± 21.8	37.7	3.9	3.6	55.7 ± 17.0	52.4	4.2	4.0
AGENS-SemOnly	29.2 ± 5.9	66.1	2.6	2.4	96.6 ± 19.9	38.3	2.2	2.0	51.9 ± 23.4	54.3	2.2	1.9
Greedy-Replan	59.5 ± 11.7*	46.5	1.8	2.5	75.0 ± 18.5	28.1	2.1	2.2	54.2 ± 15.6	43.0	1.8	2.0
Beam-Replan	59.4 ± 11.8*	47.6	1.8	2.5	74.2 ± 18.0	27.9	2.1	2.3	54.4 ± 15.6	43.6	1.8	2.0
VILA	30.9 ± 7.1	66.2	2.7	2.5	102.4 ± 18.6	38.3	2.3	2.1	59.0 ± 22.1	51.0	2.4	2.0
Oracle	0.0 ± 0.0	93.8	49.6	100.0	0.0 ± 0.0	82.4	58.1	100.0	0.0 ± 0.0	94.5	58.9	100.0

TABLE 5
Ablations of VILA: cumulative regret (mean ± 95% CI), satisfaction (%), switches per 100 cycles and executed-violation rate (% of cycles). *: differs from VILA after Holm correction over the ablations.

	Test set A				Test set B				Test set C			
Variant	Regret	Sat.	Sw.	Viol.	Regret	Sat.	Sw.	Viol.	Regret	Sat.	Sw.	Viol.
VILA	30.9 ± 7.1	66.2	2.7	0.00	102.4 ± 18.6	38.3	2.3	0.00	59.0 ± 22.1	51.0	2.4	0.00
VILA-noMem	28.3 ± 6.3	68.2	2.8	0.00	92.4 ± 21.4	40.5	2.4	0.00	52.4 ± 20.1	56.0	2.3	0.00
VILA-Fixed	32.1 ± 5.8	66.8	7.6	0.00	85.7 ± 19.1	43.1	8.7	0.00	36.3 ± 10.0	61.2	8.0	0.00
VILA-noCooldown	33.8 ± 8.7	66.9	3.0	0.00	112.2 ± 21.7	33.5	2.7	0.00	61.8 ± 21.4	46.3	3.0	0.00
VILA-noInvalidate	30.7 ± 7.6	65.7	2.8	0.00	91.9 ± 21.4	42.2	3.1	0.00	45.5 ± 17.3	54.8	2.7	0.00
VILA-Gated	30.9 ± 7.1	66.2	2.7	0.00	103.0 ± 18.5	38.0	2.3	0.00	58.9 ± 22.1	51.0	2.4	0.00
VILA-PostHoc	30.9 ± 7.1	66.2	2.7	0.00	99.9 ± 18.7	38.5	2.3	0.00	59.0 ± 22.1	51.0	2.4	0.00
VILA-Project	30.9 ± 7.1	66.2	2.7	0.00	100.7 ± 18.5	38.3	2.3	0.00	57.5 ± 21.6	51.3	2.5	0.00
VILA-Unguarded	30.9 ± 6.8	66.1	2.4	0.78	99.9 ± 18.7	38.6	2.0	0.27	57.7 ± 21.8	51.0	2.1	1.18

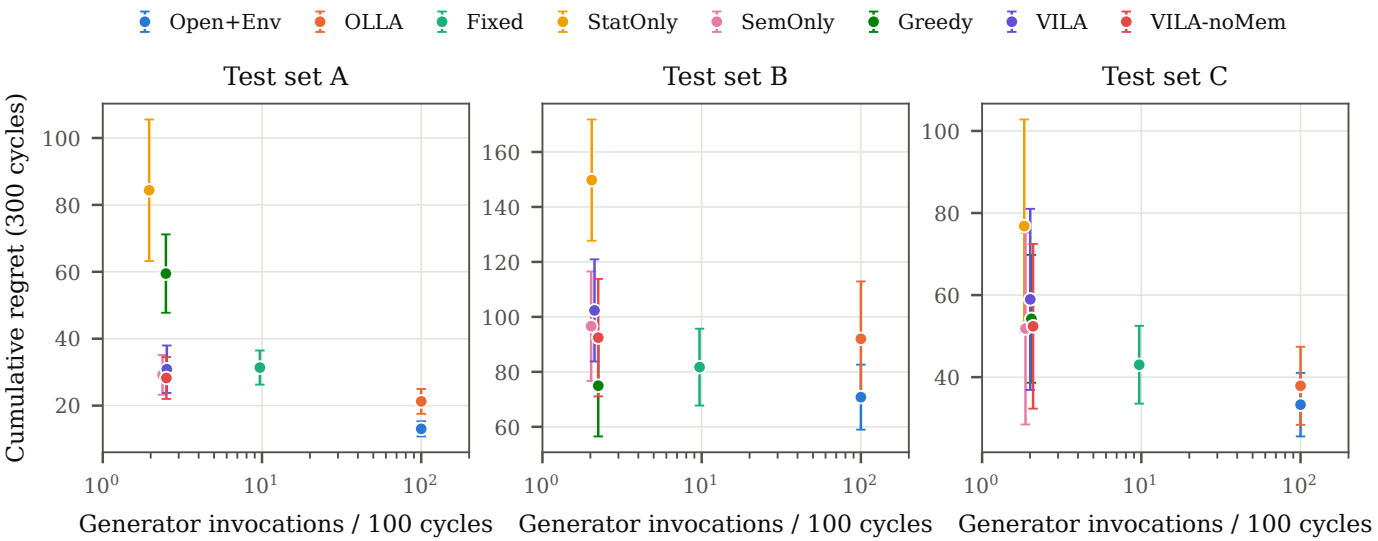


Fig. 2. Regret versus generator invocations per 100 cycles (log scale; mean ± 95% CI over 20 episodes). For Greedy-Replan, re-planning events are counted. AGENS-Open (identical to Open+Env up to the envelope), Stat+Rule and Beam-Replan are omitted for legibility; see Table 4.

each re-plan costs 27 (greedy) or 54 (beam) link evaluations, which AGENS reports as 308 s and 446 s for the full Sionna chain [15]. On B, VILA is also worse than AGENS-Fixed (Holm $p = 0.022$). The intent-detection failure under paraphrase shift described above means that intent changes are frequently not acted upon until a channel event forces a re-plan, whereas the fixed schedule re-plans regardless. Measured against the true cause, the reflection module's attributions in event windows are correct in 71.7% (A), 67.6% (B) and 61.2% (C) of 540–600 calls. All three values are above the 60% go/no-go level of the project plan, but they show a clear decline under shift.

Test set C is the hardest for event-triggered control. Vague intents, quick reversals and power-cap episodes produce many situations that call for re-planning but are not *events* in the calibrated sense. VILA-Fixed (36.3) is better

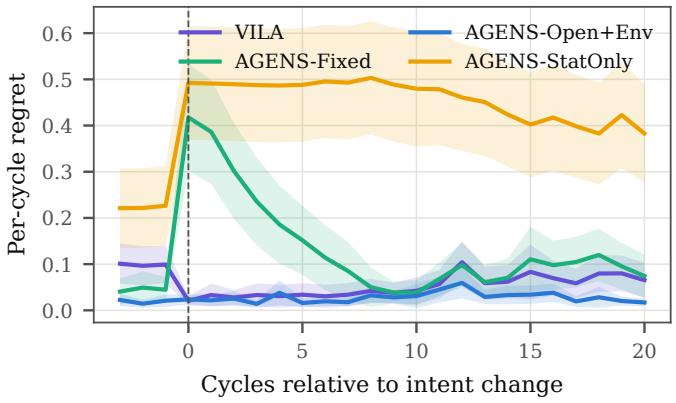


Fig. 3. Per-cycle regret aligned on the 75 intent changes of test set A with a complete $[-3, +20]$-cycle window (mean, shaded 95% CI).

than VILA (59.0; Holm $p = 0.051$), and AGENS-Open remains best (33.4).

6.4 E4: Safety Envelope (H3)

Verification. The exhaustive checks of Section 4.4 (105,840 projection checks and 125,832 decoding checks over 14 envelope states) all pass, in 134 s on one CPU core. The admissible set ranges from 372 configurations (cap 0 dB, no covariance) to 3906. No executed configuration of any guarded method violated C1–C4 in any of the $3 \times 20 \times 300$ test cycles. This is guaranteed by construction and was also asserted at run time.

What the envelope prevents. Without the envelope, AGENS-Open executes an inadmissible configuration in 0.08% (B) and 0.32% (C) of cycles, mostly through MCS-table (C1) and power-cap (C3) violations. VILA-Unguarded does so in 0.78% (A), 0.27% (B) and 1.18% (C) of cycles. The audit log shows that most of these violations come from *held* configurations rather than fresh generations. On C, 69 of 73 violation events are held configurations that became inadmissible: LMMSE estimation right after a regime change (C4), or a power offset above a newly lowered cap (C3). An envelope that only checks generator outputs would miss them. In VILA, the same situations trigger 32 re-projections of held configurations and 4 masked corrections at generation time on test set C.

Task cost of the envelope. Constrained decoding is pre-registered as equivalent to the unconstrained-then-filtered variant (VILA-PostHoc) within $\pm 2\%$ of mean reward. On test sets A and C, the two produce identical decisions in every episode. On B, the mean difference is -0.008 with a 90% CI of $[-0.022, 0.006]$, wider than the ± 0.0094 margin, so equivalence is not established there ($p_{\mathrm{TOST}} = 0.43$). Pooled over the 60 episodes, the difference is -0.0027 with a 90% CI of $[-0.0072, 0.0018]$ inside ± 0.0126 ($p_{\mathrm{TOST}} = 2.4 \times 10^{-4}$). **H3 is supported for the pooled test and for A and C, and inconclusive on B.** Unconstrained decoding followed by projection (VILA-Project) and no envelope at all (VILA-Unguarded) also differ from VILA by less than 1% of reward, with no significant difference after correction. The envelope is thus essentially free in task terms; its value lies in the guarantee.

TABLE 6
Backbone cost per cycle from (10) with measured invocation rates and context lengths (GPT-2 Medium, $n_g = 24$ decoded attribution tokens), in GFLOP and relative to AGENS-Open.

$2L$	Set	AGENS-Open	VILA (full)	VILA-Gated	Gated, cls. head
64	A	39.1	185.0 (4.74×)	105.9 (2.71×)	97.2 (2.49×)
64	B	39.1	111.2 (2.85×)	88.4 (2.26×)	76.6 (1.96×)
64	C	39.1	118.8 (3.04×)	87.7 (2.24×)	76.7 (1.96×)
256	A	161.1	188.1 (1.17×)	109.0 (0.68×)	100.3 (0.62×)
256	B	161.1	113.8 (0.71×)	91.0 (0.56×)	79.2 (0.49×)
256	C	161.1	121.3 (0.75×)	90.1 (0.56×)	79.1 (0.49×)

6.5 E5: Computational Cost and Scalability

Invocation rates and backbone compute. Table 6 combines the measured invocation rates and context lengths with the cost model (10). Because the AGENS paper does not state the number of backbone positions of the generator, we report two values, $2L = 64$ and $2L = 256$. With a reflection pass in every cycle, VILA costs $4.7\times$ the backbone FLOPs of AGENS-Open per cycle on A when $2L = 64$, and $1.17\times$ when $2L = 256$, even though it invokes the generator 40 times less often. The reflection prompt (268 tokens on average on A, at most 459), not the generator, dominates the cost. The pre-screen gate (VILA-Gated) leaves regret, triggers and switching unchanged on all three sets (Table 5) and skips 43% of reflection calls on A, but only 21–26% on B and C, where failed slots are frequent. With the classification-head reflection used in our experiments (prefill only, no decoding), the ratio drops to 2.0–$2.5\times$ for $2L = 64$ and to 0.49–$0.62\times$ for $2L = 256$. The non-LLM components are negligible: on one CPU thread, constrained decoding takes 135 μs, a projection 342 μs, a CUSUM update 24 μs and a BOCPD update 80 μs.

Inference latency (exploratory). The analyses so far assume that a generated configuration takes effect in the cycle in which it is requested. This analysis was not pre-registered; we added it after E1. AGENS reports 0.167 s per inference [15], i.e., 33 cycles of 5 ms. We therefore re-ran test sets A and B with decisions taking effect $D \in \{0, 2, 6, 33\}$ cycles after they are requested, with AGENS-Open pipelined so that a decision based on cycle-t observations is applied at $t + D$ (Fig. 6). The advantage of every-cycle generation shrinks quickly with D. On A, AGENS-Open's regret grows from 13.1 to 38.8 at $D = 6$ and to 85.3 at $D = 33$, whereas VILA's grows from 30.9 to 43.7 and 96.1. At $D = 33$ the confidence intervals of all four methods overlap. Sustaining AGENS-Open at that latency requires about 34 concurrent generator instances, against fewer than 1 for VILA (1.4 invocations per 100 cycles).

Memory budget. The bound (5) was never exceeded. The largest connector input observed was 459 tokens, against a bound of 520 and a context of 1024. The recursive summary of a 300-cycle episode needs only 93 tokens on average, so budgets above 100 tokens make no difference (Fig. 7). Fidelity probes, whose answers are checked against the uncompressed memory (number of intent changes, previous intent, dominant failure mode), remain at 100% down to $B_s = 40$ tokens. The last-switch probe, checked against the simulator ground truth, reaches at most 55% and drops to 35% at $B_s = 40$. Its ceiling is set by missed or mis-

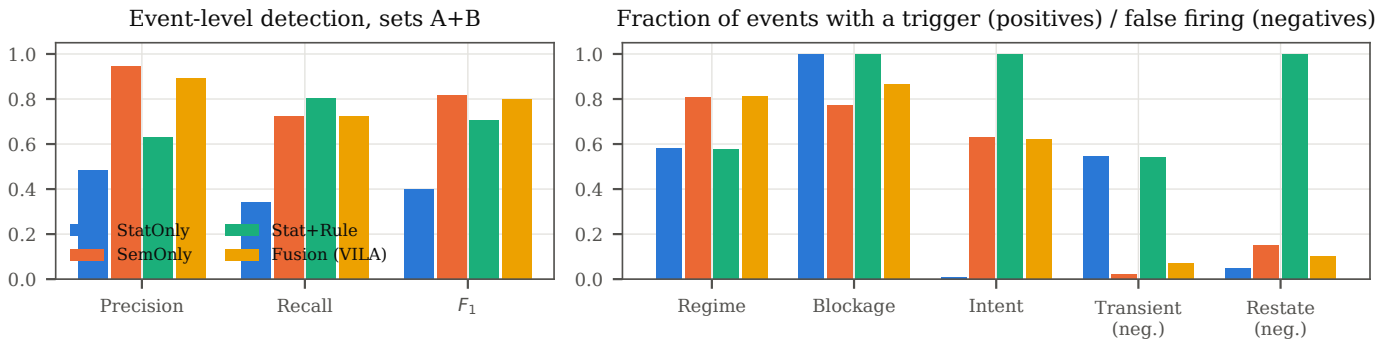


Fig. 4. Trigger quality on test sets A+B. Left: event-level precision, recall and F_1. Right: detection rate per positive event type, and firing rate on negative events (transient fades; paraphrased restatements, measured on set C).

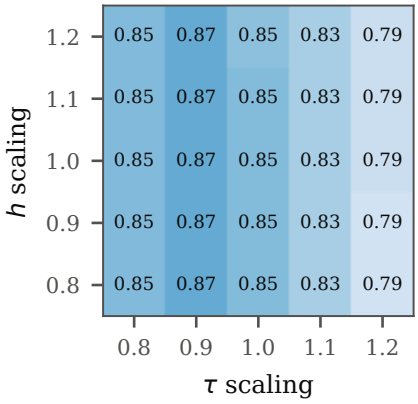


Fig. 5. Event-level F_1 of VILA on test set A (10 episodes) when the calibrated CUSUM threshold h and fusion threshold τ are scaled by 0.8–1.2 (τ capped at 0.999).

TABLE 7
Outcome of the pre-registered hypotheses.

Hyp.	Result	Verdict
H1	Intent-window regret -66.8% vs. AGENS-Fixed (CI 56.8–75.3%), switching 2.7 vs. 7.4/100. Gain attributable to triggering; memory ablation n.s.; whole-episode regret equal.	Supported as specified; mechanism differs
H2	Fusion $F_1 = 0.80$ vs. semantic-only 0.82 (-2.2%); FPR 0.30%.	Not supported
H3	Pooled TOST $p = 2.4 \times 10^{-4}$ (margin 2%); identical on A, C; B inconclusive.	Supported (pooled)

attributed regime switches, not by compression. Likewise, the intent trajectory recorded in memory matches the true number of intent changes in only 52.5% of episodes, because the intent posterior is noisy on vague and paraphrased requests. Memory is therefore limited by perception, not by compression.

7 Discussion

What the evidence supports. Three findings are robust across test sets and analyses. (i) A table-driven safety envelope with prefix-completable decoding provides a mechanically verified admissibility guarantee at negligible task cost. It also catches a failure mode that an output-only check misses: held configurations invalidated by envelope changes. These make up most of the violations of an unguarded controller. (ii) Event-triggered re-planning reduces generator invocations by a factor of 40 and configuration switches by a factor of 15 relative to every-cycle generation. It responds to intent changes within the same cycle, where a fixed schedule responds within half a period. (iii) In our simulator, the statistical change detectors add little to a learned semantic deviation score. They detect blockages well but are blind to intent changes and cannot tell transient fades from regime onsets, which matches the motivation for Q2 but not the fusion gain anticipated by H2.

What it does not support. The numeric memory view did not improve decisions (Q1), and cause-aware invalidation was neutral to harmful. The fused trigger did not beat the semantic-only trigger. When generation is free and instantaneous, per-cycle regeneration attains the lowest regret; VILA's advantage is in compute, switching and auditability, not in regret. Under distribution shift, the learned components of VILA degrade more than model-based search, mainly because intent changes phrased in unseen ways go unnoticed. Most of these negative results trace back to perception: an intent encoder that does not generalise to paraphrases, and a memory whose content is only as good as the attributions written into it. They are therefore specific to the compact backbone. A GPT-2 Medium reflection module may well recognise paraphrased intent changes; whether it does is an empirical question that our results neither answer nor pre-empt.

Threats to validity. *Construct validity.* The regret is defined against a clairvoyant per-cycle oracle that knows the hidden interference. Event-triggered controllers, which hold configurations by design, are penalised for every cycle-level fluctuation, so the regret overstates the cost of holding relative to a latency-aware oracle. The reward inherits the AGENS normalisations, including our intent-conditioned BER reference. *Internal validity.* Two design decisions were taken after inspecting training and calibration data, before any test set was run: the removal of two leaky reflection inputs, and closed-loop threshold calibration. The latency analysis was added after the main results and is labelled exploratory. All methods share episodes and random numbers, and all comparisons are paired. *External validity.* The link-abstraction surrogate reproduces the structure of the AGENS chain but not its exact numbers; event rates are

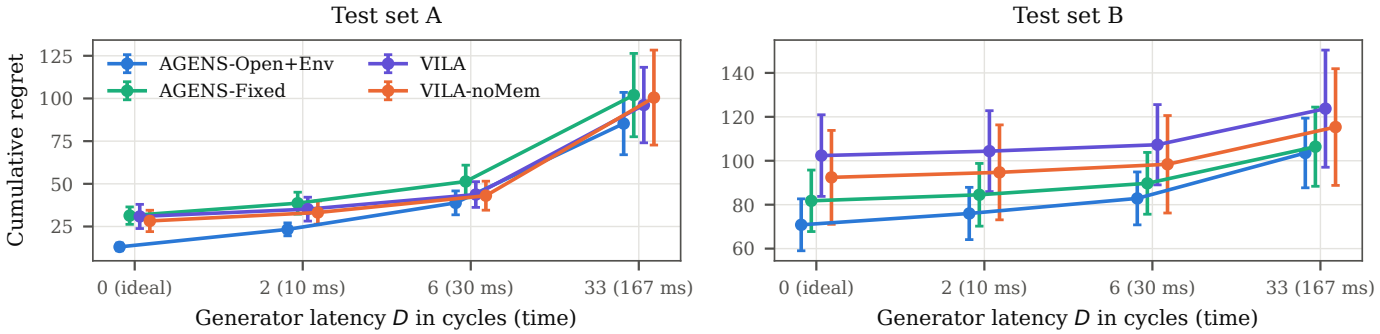


Fig. 6. Cumulative regret when generated configurations take effect D cycles after the request (exploratory analysis; mean ± 95% CI over 20 episodes). $D = 33$ corresponds to the 0.167 s inference time reported for AGENS.

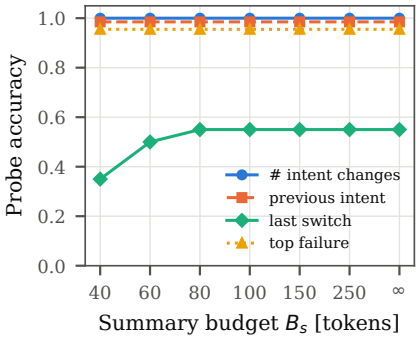


Fig. 7. Memory probe accuracy versus summary budget B_s (40 episodes, sets A and C). The three fidelity probes coincide at 1.0 and are offset slightly for visibility.

higher than in most real sessions; and only one link is controlled. The training and calibration sets are disjoint from the test sets, but all three come from the same simulator family.

Implications for AI-native air interfaces. The results suggest a division of labour for 6G link control. A heavy intent-conditioned generator should be invoked only on events, inside a verifiable envelope, while fast variations are left to cheap inner loops, as OLLA and slot-level precoding already handle them today. The envelope is the component most ready for standardisation discussions. Its constraint tables are explicit, its guarantee can be checked exhaustively, and its audit log gives operators a record of every corrected decision. The reflection module is the most expensive component (Table 6). It must be gated, distilled into a classification head, or run on a smaller model before an LLM-based reflection is affordable at the near-real-time RIC timescale [44].

Reproducing the LLM instantiation. Replacing the compact networks by the AGENS GPT-2 Medium generator and a rank-8 LoRA reflection adapter [12] requires no change to the trigger, the envelope, the simulator or the statistics. Every number in Tables 4–7 can be regenerated with the released scripts once the two model files are replaced. Coupling the controller with full Sionna link runs, ns-3 [53] or OpenAirInterface [54] is the natural route to closing the simulation-to-reality gap.

8 Conclusion

This paper extended intent-aware single-step PHY configuration generation to sustained, closed-loop operation. VILA wraps an AGENS-style generator with a token-budgeted semantic memory, a calibrated dual-signal re-planning trigger with a cooldown, and a table-driven safety envelope whose prefix-completable decoding and projection are proven sound and checked exhaustively. In 60 test episodes (18,000 cycles) over in-distribution, shifted and adversarial conditions, the envelope removed all protocol violations at no measurable task cost. Event-triggered re-planning cut generator invocations by a factor of 40 and switching by a factor of 15 relative to every-cycle generation, and reduced post-intent-change regret by two-thirds relative to fixed-period re-planning. The memory and the statistical-semantic fusion, on the other hand, did not deliver the gains we had hypothesised with the compact backbone used here, and every-cycle generation retained the lowest regret whenever its inference cost and latency were ignored. Future work will repeat the study with the GPT-2 Medium generator and LoRA reflection, add a latency-aware oracle, couple the event-level agent with fast inner loops, and validate the closed loop on full link-level and over-the-air platforms.

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Data and Code Availability

The closed-loop simulator, envelope tables and verification test, training and calibration scripts, trained model files and all per-episode logs are released at [repository URL to be inserted]. The GPT-2 BPE vocabulary is used only for token counting.

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